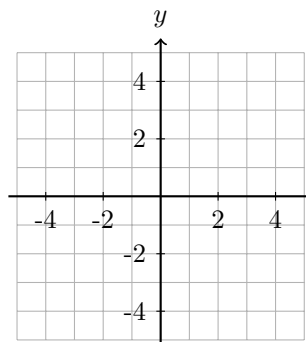


Algebra 2 - Unit 6 Bellwork: Week 3

Name: _____ Date: _____ Period: _____

Day 10: Review - Mixed Graphing + Prior Knowledge

1. Graph $f(x) = \sqrt[3]{x+1} - 2$



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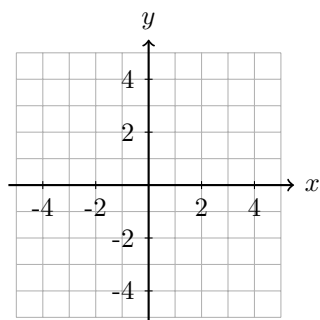
2. If $f(x) = x + 5$, what input gives output 12?
(Hint: Find $f^{-1}(12)$)

Answer:

3. $f(x) = 2x$ and $g(x) = \frac{x}{2}$ are inverses.

Verify: $f(g(4)) =$

4. [SAT] Which system is represented by the region below $y = x + 2$ and above $y = -x$?



A) $y < x + 2$ and $y > -x$

B) $y > x + 2$ and $y < -x$

Answer:

Day 11: Review - Finding Inverse Functions

4 Steps: (1) Replace $f(x)$ with y (2) Swap x and y (3) Solve for y (4) Write as $f^{-1}(x)$

1. Find $f^{-1}(x)$ for $f(x) = x - 7$

$f^{-1}(x) =$

2. Find $f^{-1}(x)$ for $f(x) = x^3 + 2$

$f^{-1}(x) =$

3. Find $f^{-1}(x)$ for $f(x) = \sqrt[3]{x-4}$

$f^{-1}(x) =$

4. [SAT] If $f(x) = 3x + b$ and $f(4) = 17$, what is b ?

A) 3 B) 5 C) 7 D) 9

Answer:

Day 12: Review - Verifying Inverses

1. If $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$, find $f(g(x))$.
What does this tell you about f and g ?
 $f(g(x)) =$

What does this tell you?

2. Find $f^{-1}(x)$ for $f(x) = (x + 3)^3$

$$f^{-1}(x) = \text{$$

3. Solve: $2\sqrt[3]{x - 5} = 6$

$$x = \text{$$

4. [SAT] The table shows a linear function.

x	0	2	4	6
y	-3	1	5	9

What is the y -intercept?

A) -3 B) 0 C) 2 D) 5

Answer: